

“Fakeflix” Testing

EN.601.422/622 Software Testing and Debugging

Spring 2025

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01

SUT Introduction

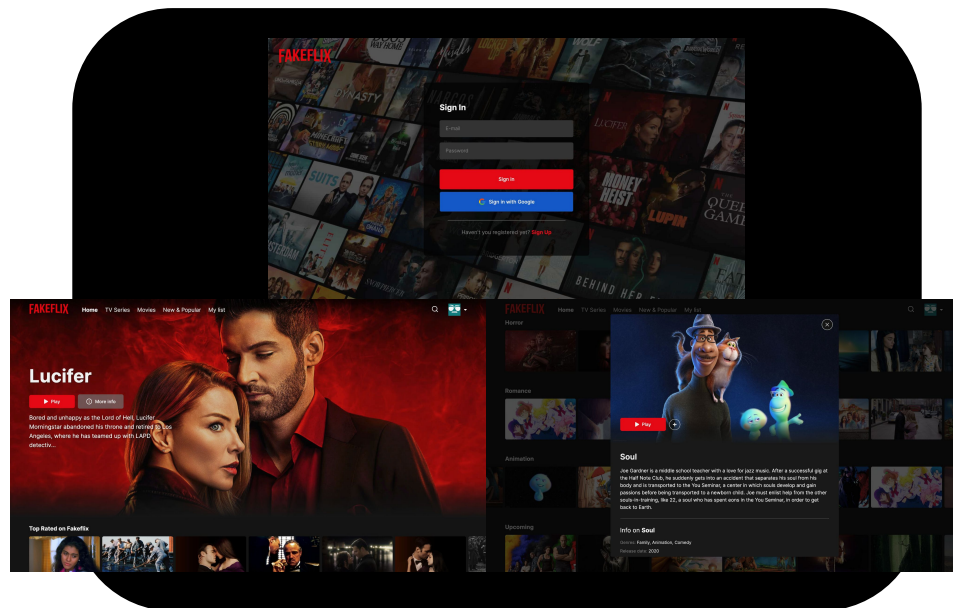
Fakeflix

Link: <https://github.com/Th3Wall/Fakeflix>

Fakeflix is a visually rich, responsive **clone of Netflix**. It mimics core Netflix functionalities such as user login, browsing content, watching trailers, and dynamic recommendations.

Codebase Metrics

- Language: JavaScript (React, Node.js)
- Frontend: 26 JSX files, 107 lines of blank, 2 lines of comment, 1562 lines of code.
- Backend Modules: 65 JavaScript files, 359 lines of blank, 26 lines of comment, 3020 lines of code.



Fakeflix Intro

01 What is the SUT app you have chosen and what is it used for?

SUT: Fakeflix
Used for: mimics core
Netflix functionalities

02 What kind of an app is it?

Web-app

03 On what platform(s) is it run?

Used on web browsers:

- optimized for desktop usage
- supports responsive layouts for mobile devices

04 Who are the potential/active users of the app

developers and students
interested in learning
full-stack development

05 what technologies and programming language(s) the app is implemented/developed with?

Frontend:

- React.js(JavaScript) for building UI
- Redux for state management
- Axios for making HTTP requests

Backend: Firebase

Authentication: JWT-based

Data Fetching: REST APIs

02

Live Showcase

White-box Testing Result

```
1  import { moviesActionTypes } from './movies.types';
2
3  const initialState = {
4    ...loading: false,
5    ...error: '',
6    ...data: []
7  }
8
9  const topRatedMoviesReducer = (state = initialState, {type, payload}) => {
10    ...switch (type) {
11      ...case moviesActionTypes.FETCH_TOP_RATED_MOVIES_REQUEST:
12        ...return {
13          ...state,
14          ...loading: true
15        }
16      ...case moviesActionTypes.FETCH_TOP_RATED_MOVIES_SUCCESS:
17        ...return {
18          ...state,
19          ...data: payload,
20          ...loading: false,
21          ...error: ''
22        }
23    }
```

Redux global state management:

- Action
- **Reducer**
- **Selector**
- Store
- Provider
- Dispatch

White-box Testing Result

```
1 import topRatedMoviesReducer from "../topRated.reducer";
2 import { moviesActionTypes } from "../movies.types";
3
4 describe("topRatedMoviesReducer", () => {
5   const initialState = {
6     loading: false,
7     error: "",
8     data: [],
9   };
10
11   it("should ignore unknown action and return unmodified state", () => {
12     const next = topRatedMoviesReducer(initialState, {
13       type: "Some unknown type",
14     });
15     expect(next).toEqual(initialState);
16   });
17
18   it("should handle FETCH_TOP_RATED_MOVIES_REQUEST", () => {
19     const action = { type: moviesActionTypes.FETCH_TOP_RATED_MOVIES_REQUEST };
20     const next = topRatedMoviesReducer(initialState, action);
21     expect(next).toEqual({
22       ...initialState,
23       loading: true,
24     });
25   });
26 });
```

Reducers are pure functions comprised mainly conditional statements.

They take two arguments, previous state, and action, to produce a new state.

Test cases must cover all possible branches and input partitions.

Black-box Testing Result

```
> fakeflix@0.1.0 test:unit:report
> vitest run --coverage

Loaded vitest@3.1.2 and @vitest/coverage-v8@3.1.1.
Running mixed versions is not supported and may lead into bugs
Update your dependencies and make sure the versions match.

RUN v3.1.2 C:/Users/FrankF/Desktop/Debugging/FinalProject/Fakeflix
Coverage enabled with v8

> src/unit-tests/hooks.blackbox.test.tsx (11 tests | 3 failed) 47ms
  ✓ EP: returns genre names for valid IDs 10ms
  ✓ EP: returns empty for unknown IDs 1ms
  ✓ BA: edge case with empty input 1ms
  ✓ EP: image renders with useLazyLoad ref 11ms
  ✓ EP: click outside triggers onClose 3ms
  ✓ EP: click inside does NOT trigger onClose 3ms
  ✗ EG: returns items for known category 11ms
    → Cannot read properties of null (reading 'filter')
  ✗ EG: handles unknown category 2ms
    → Cannot read properties of null (reading 'filter')
  ✗ EG: retrieves data for valid type 2ms
    → Cannot read properties of null (reading 'map')
  ✓ BA: scroll detection returns boolean 1ms
  ✓ EP: returns width and height of viewport 1ms
```

Out of 11 blackbox hook tests, 8 passed and 3 failed due to `null` access errors on missing Redux state (`selectedConfigArray`), working on the fix and adding more tests.

The failed tests involve category and data retrieval logic that require mocking proper state structure.

GUI Testing

```
15 failed
  [auth-tests] > auth.spec.js:80:5 > Authentication Page Tests > Sign Up Navigation > should navigate to sign up form when clicking
the Sign up link
  [auth-tests] > auth.spec.js:132:5 > Authentication Page Tests > Sign Up Navigation > should navigate back to login form when click
ing the Sign in link
[chromium] > home.spec.js:24:5 > Home Page Tests > Banner Section > should display the featured banner
[chromium] > movie.spec.js:26:5 > Movies Page Tests > Movies row tests > should display rows with correct titles
[chromium] > myList.spec.js:25:9 > My List Page Tests > Routing to my list page > should be on my list page
[chromium] > popular.spec.js:26:5 > News & Popular Page Tests > News & Popular row tests > should display rows with correct titles

[chromium] > tvseries.spec.js:26:5 > TV Series Page Tests > TV servies row tests > should display rows with correct titles
[firefox] > movie.spec.js:26:5 > Movies Page Tests > Movies row tests > should display rows with correct titles
[firefox] > myList.spec.js:25:9 > My List Page Tests > Routing to my list page > should be on my list page
[firefox] > popular.spec.js:26:5 > News & Popular Page Tests > News & Popular row tests > should display rows with correct titles
[firefox] > tvseries.spec.js:26:5 > TV Series Page Tests > TV servies row tests > should display rows with correct titles
[webkit] > movie.spec.js:26:5 > Movies Page Tests > Movies row tests > should display rows with correct titles
[webkit] > myList.spec.js:25:9 > My List Page Tests > Routing to my list page > should be on my list page
[webkit] > popular.spec.js:26:5 > News & Popular Page Tests > News & Popular row tests > should display rows with correct titles
[webkit] > tvseries.spec.js:26:5 > TV Series Page Tests > TV servies row tests > should display rows with correct titles

2 flaky
[chromium] > navbar.spec.js:25:9 > Navbar Tests > Navbar logo > should display the correct logo
[chromium] > navbar.spec.js:84:9 > Navbar Tests > Active link highlighting > should add activeNavLink class to clicked navbar link

11 skipped
45 passed (1.2m)

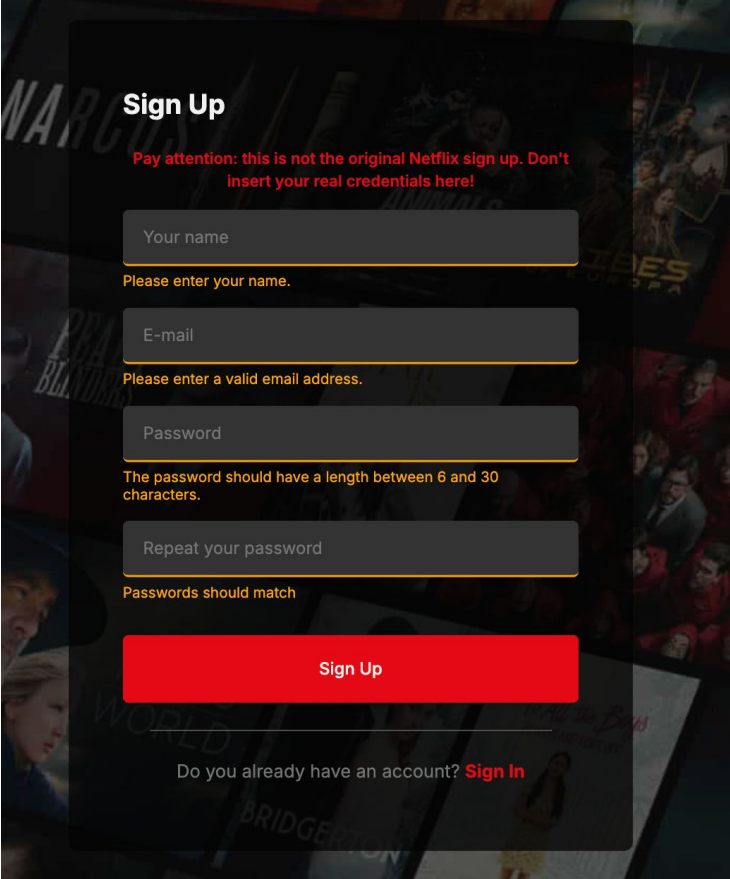
Serving HTML report at http://localhost:9323. Press Ctrl+C to quit.
```

Q	All 73	Passed 45	Failed 15	Flaky 2	Skipped 11
4/20/2025, 9:19:44 PM Total time: 39.9s					
▼ movie.spec.js					
✖ Movies Page Tests > Movies row tests > should display rows with correct titles chromium					35ms
movie.spec.js:26					
✖ Movies Page Tests > Movies row tests > should display rows with correct titles firefox					663ms
movie.spec.js:26					
✖ Movies Page Tests > Movies row tests > should display rows with correct titles webkit					1.5s
movie.spec.js:26					
▼ myList.spec.js					
✖ My List Page Tests > Routing to my list page > should be on my list page chromium					1.5s
myList.spec.js:25					
✖ My List Page Tests > Routing to my list page > should be on my list page firefox					276ms
myList.spec.js:25					
✖ My List Page Tests > Routing to my list page > should be on my list page webkit					158ms
myList.spec.js:25					
My List Page Tests > should display no items in my list chromium					0
myList.spec.js:30					
My List Page Tests > should display no items in my list firefox					0
myList.spec.js:30					
My List Page Tests > should display no items in my list webkit					0
myList.spec.js:30					
▼ popular.spec.js					
✖ News & Popular Page Tests > News & Popular row tests > should display rows with correct titles chromium					514ms
popular.spec.js:26					
✖ News & Popular Page Tests > News & Popular row tests > should display rows with correct titles firefox					364ms
popular.spec.js:26					
✖ News & Popular Page Tests > News & Popular row tests > should display rows with correct titles webkit					221ms
popular.spec.js:26					
▼ tvseries.spec.js					
✖ TV Series Page Tests > TV series row tests > should display rows with correct titles chromium					25ms
tvseries.spec.js:26					

GUI Testing

Test more scenario that needs some trigger

For example, if there is nothing input when signing up, whether the validation message can be correctly prompted



Sign Up

Pay attention: this is not the original Netflix sign up. Don't insert your real credentials here!

Your name

Please enter your name.

E-mail

Please enter a valid email address.

Password

The password should have a length between 6 and 30 characters.

Repeat your password

Passwords should match

Sign Up

Do you already have an account? [Sign In](#)

03

Summarization & Q&A

Summary

Our testing is in progress...

SUT: Fakeflix – a Netflix clone for practicing full-stack development

White-box Testing: Focused on Redux reducers to ensure internal logic correctness

Black-box Testing: Validated system behavior through UI-level tests, including event handling and rendering behavior

Integration Testing & GUI testing: Ensure elements can be seen on the screen and can be interacted correctly

Challenges: Mocking Firebase dependencies and ensuring state coverage without modifying core app logic

Outcome: Built a robust suite of automated tests with both logic-level and user-level validation